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Crossing the Border: A Difference-in-Differences Evaluation of Decontribuzione Sud

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Abstract

Decontribuzione Sud is a place-based payroll tax cut introduced in Southern Italy in 2020 to support employment in response to the COVID-19 pandemic.

This paper assesses the causal effect of Decontribuzione Sud by comparing firms in Southern and Central Italian provinces on either side of the policy border, using a Difference-in-Differences approach.

The estimates indicate an overall positive impact: three years after the introduction, employment increases by 8.3%, value added by 11.3%, and return on assets by 1.47 percentage points. However, these profitability gains do not primarily reflect efficiency improvements, as value added per employee remains unchanged, but rather stem from the mechanical effect of labour cost savings.

Aggregating at the municipality level, the increase in aggregate employment income is driven almost entirely by employment growth rather than by a pass-through of the benefit to incumbent workers.

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1 Introduction

Decontribuzione Sud is a contribution relief introduced in the Italian legal framework to limit the negative impact of COVID-19 on employment in Southern Italy. Over time, this measure has evolved into an instrument aimed at reducing the structural territorial disparities across the country. The policy provides employers located in Southern regions with a substantial reduction in social security contribution.

While most of the current literature on payroll tax cuts focuses on employment and wages, this thesis provides a comprehensive analysis of firm-level accounting outcomes. Specifically, it investigates the policy's impact on firm survival, investments, productivity, and profitability by analysing balance sheet data.

To identify the causal impact of the intervention, the empirical strategy employs a Difference-in-Differences (DiD) approach. The model exploits the eligibility discontinuity between Southern and Central regions by comparing only firms located in provinces close to this border before and after the intervention.

The evidence demonstrates that Decontribuzione Sud is effective in increasing the survival probability of beneficiary firms. Eligible enterprises experience a positive scale effect, expanding both their labour force and capital investments, which translates into higher production volumes. However, this enlargement does not yield structural efficiency improvements; therefore, the observed profitability gains are mainly attributable to labour cost savings. Furthermore, on an aggregate basis, the increase in total labour income is almost fully driven by net employment creation, rather than by wage premiums captured by incumbent workers. However, the cost-benefit analysis reveals that this job creation involves a substantial fiscal burden. The estimates indicate an expenditure of approximately €25,200 per additional worker-year, implying that each extra euro of labour income generated requires roughly €1.25 in public spending.

The thesis is structured as follows: Section 2 provides a review of the relevant empirical literature on labour tax cuts; Section 3 explains the institutional background and mechanisms of Decontribuzione Sud; Section 4 presents the administrative data used in the analysis; Section 5 details the empirical methodology; Section 6 shows the main results of the firm-level analysis; Section 7 and 8 presents robustness checks and additional analyses; and Section 9 concludes by discussing the policy implications of the findings.

2 Literature Review

Hiring subsidies and payroll tax cuts have been widely used across countries as a tool to stimulate employment, yet their effectiveness and the channels through which they operate vary considerably depending on policy design. This chapter reviews the empirical literature evaluating such policies along three dimensions. First, it examines place-based schemes, which target firms or workers in specific geographic areas. Second, it turns to policies targeted at specific demographic or socioeconomic groups. Finally, the chapter examines a smaller literature that studies the impact of these policies on firms' accounting outcomes.

Starting with place-based measures, a well-established strand of literature on Nordic countries consistently shows that payroll tax cuts fail to stimulate local employment, with a substantial portion of the subsidy shifting to workers through higher wages. For instance, Bennmarker et al. (2009) evaluate a 10-percentage-point payroll tax reduction in Sweden's Regional Support Areas (RSA A). Rather than driving job creation, they find that a quarter of the labour cost reduction translated into wage increases. Similar dynamics are documented in Finland by Korkeamäki and Uusitalo (2009), who analyse a temporary 4.1% tax cut implemented in 2003 and targeted at low-employment northern municipalities. While employment remained unaffected, wages rose by 2%, implying a 50% pass-through to workers. Further corroborating this pattern, Stokke (2021) examines a 4.2-percentage-point reduction across 32 Norwegian municipalities in 2000, again finding no significant employment effects but noting a 30% wage shift in favor of employees.

By contrast, Benzarti and Harju (2021) and Ku et al. (2020) document significant negative employment effects following payroll tax increases in Finland and Norway, respectively. Benzarti and Harju (2021) exploit a quasi-experimental setting in Finland where employers faced a discontinuous tax increase of 2.6 percentage points on average if their capital depreciation exceeded a specific threshold. They find that firms above this threshold experienced a 17.7% reduction in net labour costs and a 9.1% decline in employment, with no corresponding effect on net wages. Similarly, Ku et al. (2020) evaluate an EU-mandated harmonization reform in Norway that abolished geographically differentiated rates, effectively raising payroll taxes by up to 6.6 percentage points in previously low-tax zones. Their estimates show that a 1 percentage point increase in the payroll tax rate reduces local employment by 1.37 percent and local wages by 0.32 percent, implying a partial wage pass-through of 46 percent.

In a context characterised by high labour market informality, the effects of place-based payroll tax cuts may manifest differently. Asik et al. (2022) evaluate a policy in Turkey that reduced employers' social security contributions by 6 percentage points in socioeconomically disadvantaged regions. Focusing on small firms they estimate that the subsidy increased registered employment

among beneficiary firms by 6-8%. Since administrative data only capture registered workers, they test the job creation mechanism using two alternative sources. First, using the Household labour Force Survey, they show that while the likelihood of having a formal contract increased significantly, the overall probability of being employed remained unchanged. Second, they analyse firm electricity consumption as a proxy for actual economic activity and find no break in trend in the treated provinces following the policy implementation. These findings indicate the employment increase was predominantly driven by the formalization of existing informal jobs rather than new job creation.

The evidence from these place-based interventions highlights that the impact of payroll tax changes is highly context-dependent. In some instances, tax reductions fail to stimulate job creation because the benefits are largely absorbed by incumbent workers through higher wages. Conversely, when tax rates increase, downward wage rigidities often prevent wage adjustments, forcing firms to respond through significant employment cuts. Finally, in settings characterised by a large shadow economy, these measures primarily act as a catalyst for formalizing existing jobs rather than generating new net labour demand.

According to Card et al. (2018), the empirical evidence suggests that hiring subsidies are more effective when applied to specific groups facing greater difficulty in finding a job, such as women and long-term unemployed individuals. Saez et al. (2019) and Saez et al. (2021) study the impact of a payroll tax cut for workers aged 19 to 26, introduced in Sweden in 2007. They find that the reform increased the employment rate among young workers by 2 percentage points. However, it reduced gross wages without affecting net wages, implying a 100% pass-through to firms in both the short and medium-to-long run.

Gilles et al. (2017) evaluate a French tax credit, which applied at a uniform rate to gross wages up to 2.5 times the minimum wage, thereby targeting low- and middle-wage workers, rising from 4 percent in 2013 to 6 percent in 2014. They find that firm responses varied with the intensity of benefit received: positive effects on average employment, payroll, and hours worked emerged only among firms benefiting from the maximum subsidy rate, while wage effects were positive for the highest-benefit firms but negative or insignificant for the lowest.

The success of targeted policies is not always guaranteed, however. In 2006, Finnish employers became eligible for a wage subsidy targeted at full-time, low-wage, older workers. Huttunen et al. (2013) evaluate the policy and find a positive but statistically insignificant effect on both employment (extensive margin) and working hours (intensive margin).

Turning to evidence from Italy, Rubolino (2022) evaluates an employer-borne payroll tax cut for female hires introduced in Italy in 2013. He finds that the subsidy effectively narrowed the

employment gap, generating sustained growth in female employment with no offsetting effect on male employment. However, the policy failed to reduce the gender pay gap, largely because 85.5 percent of the tax cut was passed through to firms rather than to workers.

Ciani et al. (2025) study a subsidy introduced in Italy in 1990 and in force until the end of 2014, targeting firms that hired via permanent contract individuals unemployed for at least 24 months or covered by the national short-time work compensation scheme. The subsidy covered 100% of social security contributions for 3 years in Southern Italy, versus 50% elsewhere. They find that the policy increased the weekly probability of finding a permanent job for eligible individuals in Southern Italy by 41 percent compared to the counterfactual scenario without the subsidy, but detect no effect on wages.

Overall, the evidence from policies targeting specific demographic or socioeconomic groups highlights that while these interventions successfully improve the employability of disadvantaged individuals—such as the long-term unemployed or low-wage earners—they rarely translate into wage gains. Probably because vulnerable workers typically lack bargaining power, the reduction in labour costs is almost entirely absorbed by employers.

The largest portion of hiring subsidy evaluations is limited to employment and wage effects; however, some studies also examine firms' investment, productivity, and profitability. Among the papers cited above, Saez et al. (2019) show that firms with a higher share of young workers in 2006 experienced a boost in economic activity along several dimensions, including total assets (5.8%), sales (3.1%), value added (6.1%), and profits (EBIT, 8.1%). Rubolino (2022) similarly reports that firms with a lower pre-reform share of female workers experienced increases in per-worker sales (6.4%), profits (5.3%), value added and capital (both 6.9%) following the introduction of the payroll tax cut. Benzarti and Harju (2021) find mixed results: a decrease in total investment (-14.3%) and sales (-6.8%), alongside an increase in both labour (12.8 %) and capital (11.3%) productivity. Gilles et al. (2017) detect no effect on investment, productivity, or dividends, but a positive effect on value added and gross operating surplus limited to the top quartile of beneficiaries.

Depalo and Viviano (2024) study a hiring subsidy for all new open-ended contracts, consisting of a 3-year exemption from social security contributions, capped at €8,060 per year, introduced in 2014 in Italy. They use firm-specific potential contribution savings as an instrument and identify a positive effect on open-ended hiring, a persistent reduction in labour costs, an increase in firm size, and a decline in both the capital-to-labour ratio and productivity. Santoni et al. (2025) examine the same reform and show that firms with higher wage premia capture a smaller share of the subsidy; these firms are also less productive, smaller, and employ a less experienced workforce.

Finally, the empirical literature has recently provided direct evaluations of Decontribuzione Sud, yielding contrasting results. On one hand, Albanese et al. (2026) employ a Regression Discontinuity Design on a sample of small and medium-sized enterprises (SMEs) between 2019 and 2023, finding that the policy effectively increased productivity and profitability but failed to stimulate employment or raise wages. On the other hand, a parallel evaluation by Scrutinio (2026) exploits the geographical discontinuity across bordering municipalities using a Difference-in-Differences approach and reveals a different employment dynamic. He finds that the policy increased local employment by approximately 10% after four years. Crucially, Scrutinio (2026) demonstrates that this employment expansion was driven primarily by larger establishments employing more than 50 workers, underscoring the limitations of restricting the analysis exclusively to SMEs. Nevertheless, consistent with Albanese et al. (2026), Scrutinio (2026) detects no significant impact on average wages.

Taken together, the evidence shows that the effects of hiring subsidies and payroll tax cuts depend heavily on policy design and institutional context. Place-based schemes often fail to generate employment when their incidence falls on wages, while targeted schemes improve the employability of disadvantaged groups but rarely raise their pay. The evidence on firms' accounting outcomes is thinner and mixed, and it comes mostly from targeted or economy-wide reforms rather than from geographically differentiated ones.

As a result, evidence on how a place-based payroll tax cut affects firms' investment, productivity and profitability across the full size distribution of firms remains fragmented, particularly in the Italian context. This thesis contributes to filling this gap by studying Decontribuzione Sud without restricting the sample by firm size. It analyses the policy's effects on both the micro and macro level. At firm level, it examines employment, productive factors, productivity and profitability. At municipality level, it evaluates the impact on aggregated employment income and income tax revenue.

3 Institutional Framework

In order to contain the negative effects on employment caused by the COVID-19 pandemic in areas already characterised by structural disadvantage and to protect existing employment levels, Decree-Law no. 104 of 14 August 2020, under Article 27, established an exemption equal to 30% of social security contributions in favour of private employers, applicable to both incumbent and newly hired workers.

The measure is directed only to regions that in 2018 recorded a GDP per capita below 75% of the

EU27 average, or in any case between 75% and 90%, combined with an employment rate below the national average. On this basis, the eligible territories are limited to the South and the Islands that are: Abruzzo, Basilicata, Calabria, Campania, Molise, Puglia, Sardegna, and Sicilia.

The exemption applied exclusively to contributions due to the Italian National Social Security Institute (INPS), excluding those owed to the National Institute for Insurance against Accidents at Work (INAIL). The agricultural sector, domestic employers, and financial sector firms are excluded from the measure. Specifically, the granting of the benefit is conditioned on compliance with the Temporary Framework for State aid measures to support the economy during the COVID-19 outbreak, adopted by the European Commission through its Communication of 19 March 2020. It is through this incorporation by reference to European rules that the exclusion of financial sector firms operates, functioning as an integrative element of the national regulatory framework. The 2026 Budget Law (Law no. 199/2025) subsequently clarified that insurance agencies (ATECO codes 66.22.01 to 66.22.04) are included within the scope of the measure, although the corresponding relief can only be recovered for the period from July 2022 to December 2024.

The exemption may be combined with other contribution-based incentives (such as those for hiring unemployed workers aged over 50 or disadvantaged women, Law no. 92/2012) as well as with economic incentives (such as those for hiring workers with disabilities, Law no. 68/1999, Article 13).

As originally established by Decree-Law no. 104/2020, the measure covered the period from 1 October to 31 December 2020. The 2021 Budget Law (Law no. 178/2020) subsequently extended it until 31 December 2029, with a progressive reduction in the exemption rate: 30% through 2025, 20% for 2026 and 2027, and 10% for 2028 and 2029.

However, Law no. 207/2024 (2025 Budget Law), replaced the original scheme with a considerably more selective one. First, eligibility is restricted to micro, small and medium-sized enterprises in Southern Italy, as defined by EU Recommendation 2003/361/EC, so as to concentrate public resources on smaller firms, which are regarded as more vulnerable. Second, the reform introduces an explicit employment conditionality: firms keep the benefit only if their workforce on 31 December of each year exceeds the level recorded a year earlier, thereby tying the subsidy directly to employment growth. Third, the exemption no longer covers all dependent employment but only permanent contracts and, in 2025, only those already in place on 31 December 2024, which makes the measure primarily a tool to support employment stability. Fourth, as reported in Table 1 the exemption becomes less generous: it is set at 25% of employers' social security contributions in 2025 and is gradually reduced to 15% by 2029, while monthly ceilings per worker are introduced at €145 in 2025, €125 in 2026 and 2027, €100 in 2028 and €75 in 2029. Access to the benefit also remains

subject to the general legal requirements, namely social security compliance (DURC), respect of labour and workplace safety rules, and application of national collective agreements. Taken together, these changes transform a broad-based reduction in labour costs into a targeted and selective instrument.

Table 1: Exemption rate and monthly cap per worker under the 2025 Budget Law

Year	Exemption rate (%)	Monthly cap per worker (€)
2025	25	145
2026	20	125
2027	20	125
2028	20	100
2029	15	75

Notes: The table reports the exemption rate and the monthly cap per worker per each year as established by Law no. 207/2024 and INPS Circular no. 32 of 30 January 2025.

4 Data

Two data sources are used: AIDA (Analisi Informatizzata delle Aziende Italiane), the Italian adaptation of the Orbis database provided by Bureau van Dijk (now Moody's Analytics), and the Ministry of Economy and Finance (MEF) personal income tax (IRPEF) statistics at the municipality level. AIDA provides data on firm's balance sheet and income statement. The variables allow to assess how the subsidy affects firms' labour demand, capital investments, productivity, and profitability. The MEF data provide territorial information at the municipality-level on the declared labour income and related fiscal variables. Therefore, while AIDA captures the employer-side response to the subsidy, the MEF gives the possibility to evaluate the policy's impact on the worker-side.

The analysis covers the period 2016–2023, providing four pre-treatment years (2016–2019) and four post-treatment years (2020–2023). This window allows to assess the impact of the policy and its evolution over time. The MEF data are available also for 2024, therefore they display one year more in the post-treatment period.

Three exclusion criteria are applied to the sample. First, the policy does not involve all sectors, therefore we exclude the firms that cannot benefit from the subsidy since they operate in one of the excluded sectors (agricultural, financial, public, and domestic). Second, in order to be able to compare firms before and after the policy we exclude firms which were not incorporated before 2020 and firms that were no longer operative from 2020. Third, the datasets are restricted to

provinces located along the policy eligibility border, exploiting the sharp administrative discontinuity between Southern and Central Italy and ensures that treated and control units operate in economically comparable territories. The treated provinces are Teramo, L'Aquila, Isernia, and Caserta; the control provinces are Ascoli Piceno, Rieti, Latina, and Frosinone. The province of Rome is excluded because it is the capital thus it has a specific economic dynamics not comparable with other territories. The first and the second exclusion criteria are valid only for AIDA data, therefore, MEF aggregated data internalises spillovers between firms and firm entry or exit from the market.

The firm-level data are winsorized at 1% to limit the influence of outliers. Below are presented the descriptive statistics of both datasets in 2019, which is considered as the baseline year.

Table 2 reports descriptive statistics for the AIDA balance-sheet data in 2019, the baseline year of the event study. The unit of observation is the firm (39,058 firms). The median firm has two employees, against a mean of 5.2, suggesting that the sample is dominated by micro firms. This is consistent with the structure of the Italian production system, which is composed mostly of small and micro enterprises. Variables such as fixed assets, production value and revenue display large dispersion, with standard deviations well above the means, indicating that firms in the sample differ widely in size.

Table 2: AIDA descriptive statistics in 2019

	Mean	Std. Dev.	Median	N
Employees	5.21	10.61	2.00	39058
Personnel Costs p.e. (th. Eur)	21.80	14.31	20.00	26373
Fixed Assets (th. Eur)	374.96	1187.88	32.00	39057
Fixed Asset p.e. (th. Eur)	61.79	189.47	10.75	26373
Production Value (th. Eur)	903.71	2573.96	184.00	39058
Value Added (th. Eur)	215.80	562.16	51.00	39058
Value Added p.e.(th. Eur)	35.43	34.33	28.83	26374
Revenue (th. Eur)	858.12	2470.95	167.00	39058
Profit / Loss (th. Eur)	23.27	104.98	5.00	39058
ROS (%)	4.78	10.95	4.40	30818
ROE (%)	14.77	32.93	9.78	35421
ROA (%)	4.60	18.57	3.55	39036

*Note:*The table reports municipality-level descriptive statistics for the year 2019, serving as the pre-treatment baseline. The unit of observation is the firm. Monetary values are in thousands of euros; ROS, ROE and ROA are in percentages. "p.e." denotes per employee. All variables are winsorised at 1%.

Table 3 reports descriptive statistics for the MEF personal income tax data in 2019. The unit of

observation is the municipality, and the sample comprises 541 municipalities located in the four pairs of border provinces. Municipalities are generally small: the median municipality has 1,285 taxpayers, against a mean of about 3,600, which reflects a distribution made up of many small municipalities and a few large ones. Employees account for most of the taxpayers, and employment income is the main component of taxable income, whereas self-employment plays a marginal role and is observed in 426 municipalities only. Unlike AIDA, the MEF data describe the local labour market as a whole, including self-employment and firms that do not appear in balance-sheet data.

Table 3: MEF descriptive statistics in 2019

	Mean	Std. Dev.	Median	N
N. Taxpayers	3,597.8	7,277.0	1,285.0	541
N. Taxable Income	3,420.8	6,891.5	1,238.0	541
N. Net Tax	2,504.8	5,299.5	836.0	541
N. Employees	1,921.4	4,024.2	605.0	541
N. Self-employed	43.0	110.6	13.0	426
Taxable Income (th. EUR)	60,389.9	138,366.5	18,900.2	541
Net Tax (th. EUR)	10,635.2	27,116.2	3,086.9	541
Employment Income (th. EUR)	34,306.0	78,094.5	10,423.6	541
Self-employment Income (th. EUR)	1,944.7	5,581.9	520.7	426

Note: The table reports municipal-level descriptive statistics for the year 2019, serving as the pre-treatment baseline. The unit of observation is the municipality. The prefix "N." denotes the number of individuals in a given category, while the column "N" reports the number of municipalities in the full sample. Monetary values are expressed in thousands of euros (th. EUR).

5 Method

The impact of Decontribuzione Sud is estimated through a Difference-in-Differences (DiD) model. The sample is divided into a treatment group, consisting of firms located in the eligible southern provinces, and a control group, consisting of firms located in the neighbouring ineligible central provinces. The two groups are compared before and after the introduction of the policy, and the change in the outcomes of treated firms relative to the change in the outcomes of control firms is attributed to the tax cut.

The identifying assumption of the DiD design is the parallel trends condition: in the absence of treatment, the outcomes of the two groups would have followed the same trajectory over time. This assumption is fundamental for causal identification, as it allows the control group's post-treatment path to serve as a counterfactual for what would have happened to the treated group had the

policy not been introduced. Since this counterfactual is inherently unobservable, the assumption cannot be directly tested; however, indirect evidence can be used to assess whether the two groups followed similar trends before the policy was implemented.

By restricting the analysis to provinces located on either side of the eligibility border, treated and control units operate in geographically proximate and economically comparable territories, making it plausible that they would have experienced similar dynamics in the absence of the policy. This sample construction maximize the probability to satisfy the parallel trend assumption, however geographic proximity is not sufficient and must be complemented by direct empirical evidence.

To provide an initial overview of the two groups before the policy shock, Table 4 reports the average value of each outcome for the treated and control groups in 2019, together with two-sample t-tests (allowing for unequal variances) of the difference between the two. The third column shows this difference, with its significance level, and the fourth column expresses it as a percentage of the control mean.

Treated and control firms are similar in size: employees, fixed assets and production value show no significant difference. Control firms have slightly higher personnel costs and value added per employee, with the latter difference weakly significant. Profitability shows a mixed pattern: ROS and ROE are significantly higher for treated firms, while ROA and profit/loss show no significant difference.

These differences do not threaten the identification strategy because time-invariant differences in levels are absorbed by the fixed effects, and the DiD design requires that the two groups would have followed parallel trends, not that they share the same pre-treatment levels. The evidence on parallel trends is presented in Table 5 below.

Table 4: Pre-treatment Summary Statistics and Balance (2019)

	Treated	Control	Difference (T-C)	Relative Diff. (%)
Employees	5.17	5.25	-0.09	-1.67
Personnel Costs p.e.	21.13	22.47	-1.35***	-5.99
Fixed Assets	373.42	376.53	-3.12	-0.83
Fixed Assets p.e.	61.54	62.05	-0.51	-0.82
Production Value	897.97	909.58	-11.61	-1.28
Value Added	210.99	220.72	-9.73*	-4.41
Value Added p.e.	35.06	35.80	-0.74*	-2.05
Revenue	851.10	865.30	-14.20	-1.64
Profit / Loss	23.86	22.66	1.20	5.30
ROS	4.97	4.59	0.38***	8.30
ROE	15.20	14.33	0.87**	6.09
ROA	4.60	4.60	0.01	0.14

Notes: Sample restricted to 2019. All variables are winsorised at the 1st and 99th percentiles. Relative difference is the difference as a percentage of the control mean. Two-sample t-test with unequal variances. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

While a static DiD specification captures the average treatment effect over the entire post-treatment period, it does not allow for heterogeneity in the timing and evolution of the policy's impact. To recover the year-by-year dynamics of the treatment effect, the analysis implements an event study specification.

The event study is built by constructing a set of time indicators measuring the distance of each observation from the policy implementation year (2020). These indicators are then interacted with the treatment dummy, so that a separate treatment effect is estimated for each year in the sample. The year immediately preceding the policy introduction (2019) serves as the baseline and is omitted from the regression. Consequently, all estimated coefficients are therefore interpreted as deviations in the treated-control differential relative to the baseline year, not as changes with respect to the preceding year. The formal specification is:

$$Y_{it} = \alpha + \sum_{\tau=-4,\, \tau\neq -1}^{+3} \beta_\tau \mathbf{1}\{t - t_0 = \tau\} \times Treated_i + \gamma_i + \eta_t + \varepsilon_{it}$$

Where:

- Y_{it} is the outcome variable for firm i in year t.
- $\mathbf{1}\{t - t_0 = \tau\}$ is a dummy variable indicating if the observation is τ years distant from the treatment year.
- $Treated_i$ is a dummy variable equal to 1 if the firm belongs to the eligible southern provinces,

and 0 otherwise.

- β_τ is the coefficient of interest, capturing the effect at event time τ for the treatment group with respect to the control group, relative to the baseline year (2019).
- γ_i represents firm fixed effects, absorbing all time-invariant unobserved heterogeneity across units.
- η_t represents year fixed effects, controlling for macroeconomic shocks that affect all municipalities simultaneously.
- ε_{it} is the standard error clustered at the municipality level.

Given that firms are weighted on the number of employees for the baseline year and $Treated_i$ does not account for actual beneficiaries of the tax cut but simply identifies areas where the policy could be applied, the estimated coefficients represent the weighted *intention-to-treat* effect of the hiring subsidy on eligible companies.

As already noted at the beginning of this section, since it is not possible to observe a counterfactual, the parallel trends assumption cannot be directly tested. In the event study literature, its plausibility is therefore typically assessed by examining the pre-treatment coefficients: if treated and control units were on divergent trajectories before the policy was introduced, this would cast doubt on the assumption that they would have remained on parallel paths thereafter. To this end, we test the null hypothesis that the pre-treatment interaction coefficients (for event years 2016, 2017 and 2018) are jointly equal to zero, using an F-test. The results, reported in Table 5, show that for seven of the twelve outcomes the null hypothesis is far from being rejected (p-value $>$ 0.30), including employees, fixed assets, profit/loss and the three profitability ratios. For personnel costs per employee, production value and value added, the p-values lie between 0.05 and 0.10, so the null is not rejected at the 5% level but the evidence is weaker. The null is rejected at the 5% level for ln(Revenue) (p-value = 0.023) and, marginally, for ln(Value Added p.e.) (p-value = 0.049), signalling the presence of a differential pre-treatment trend for these outcomes. Taken together, these results provide broad support for the parallel trends assumption for labour inputs, capital and profitability, while the results on revenue-related outcomes (revenue, production value and value added) should be interpreted with caution, since the pre-treatment coefficients suggest that treated and control firms may already have been diverging before 2020.

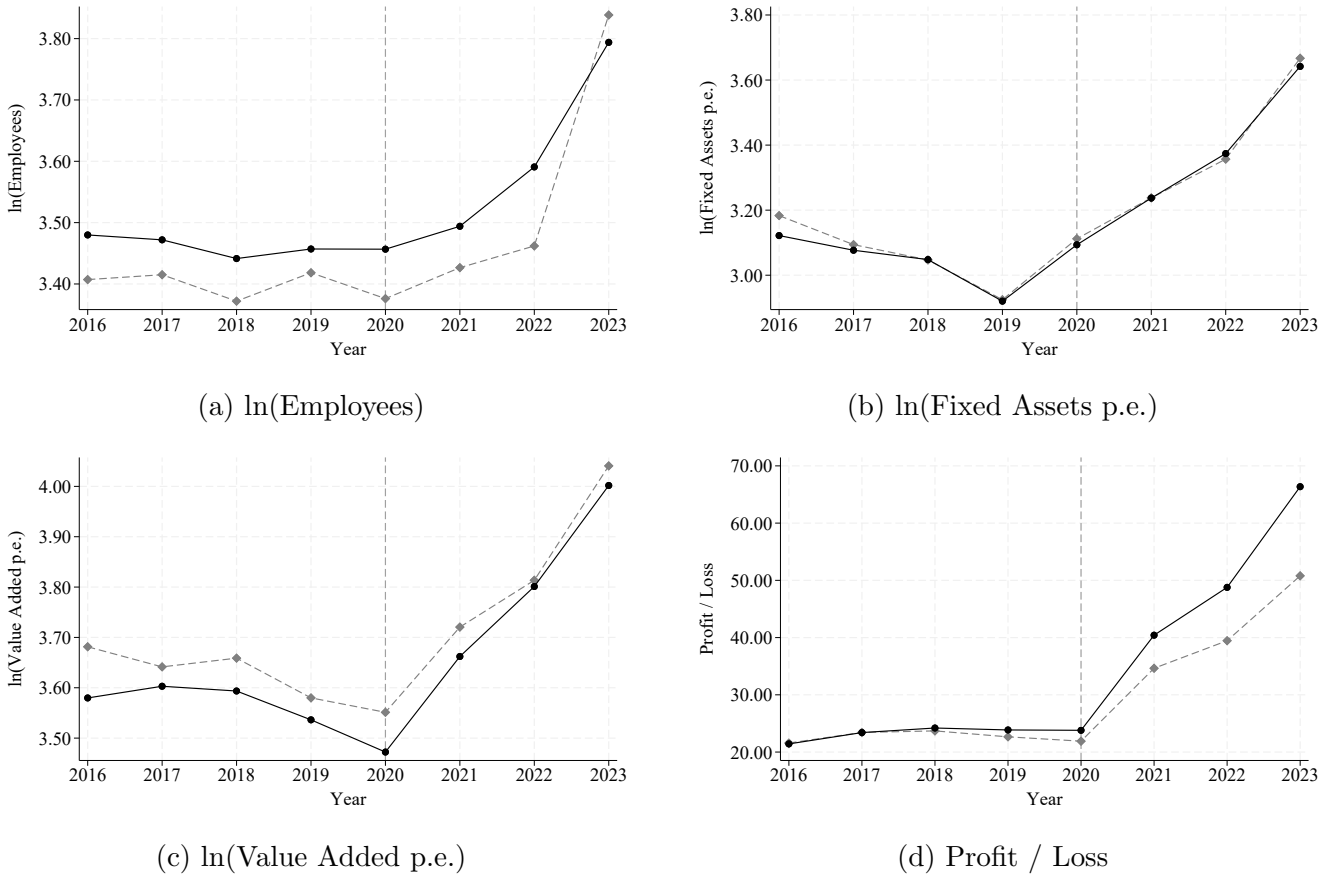
Table 5: Parallel Trends Joint F-Tests

Variable	F-statistic	P-value	Observations
ln(Employees)	0.185	0.906	166,003
ln(Personnel Costs p.e.)	2.233	0.084	163,708
ln(Fixed Assets)	0.528	0.663	175,103
ln(Fixed Assets p.e.)	0.589	0.623	158,975
ln(Production Value)	2.589	0.052	181,714
ln(Value Added)	2.162	0.092	173,943
ln(Value Added p.e.)	2.641	0.049	160,106
ln(Revenue)	3.216	0.023	179,395
Profit / Loss	1.205	0.307	267,506
ROS	0.045	0.987	167,826
ROE	0.899	0.441	167,931
ROA	0.965	0.409	184,601

Notes: Joint F-test that the pre-treatment interaction terms (2016, 2017, 2018) of the event-study specification are equal to zero. Estimates are weighted by 2019 employment, except for Profit / Loss, which is unweighted. The specification includes firm and year fixed effects. Standard errors are clustered at the municipality level.

The validity of the parallel trend assumption can be checked also by looking at the outcomes trend for each group, realised by plotting the average outcome per group in each period. Here there are the graphs for the main variables (Figure 1).

Figure 1: Parallel trends



(a) ln(Employees)

(b) ln(Fixed Assets p.e.)

(c) ln(Value Added p.e.)

(d) Profit / Loss

Note: The figure reports the trends of firms in treated and control group, and firms are weighted by 2019 employment, except for Profit / Loss, which is unweighted. The specification includes firm and year fixed effects. Standard errors are clustered at the municipality level.

6 Main Results

This section presents the results of the model previously described, the impact of Decontribuzione Sud is evaluated on both the activity status and on several balance-sheet outcomes.

6.1 Activity Status

The firm activity status is proxied by the presence of positive personnel costs, which identifies firms with at least one employee on payroll. Unlike the other firm-level outcomes presented below, which are weighted by their 2019 employment level, the estimates for the activity status are unweighted. This decision reflects the fact that when evaluating whether a business remains operational, it is reasonable to assign equal weight to all enterprises, treating each firm as an independent unit of observation regardless of its baseline size.

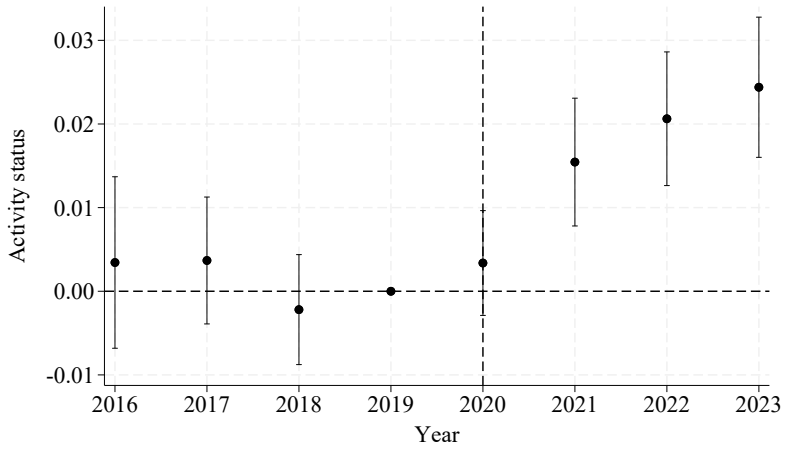
The firm activity status is proxied by the presence of positive personnel costs, which identifies firms with at least one employee on payroll.

The complete estimates are reported in Table A1 in the Appendix. The parallel trends assumption is supported by the data: the pre-treatment coefficients are jointly not significantly different from zero (F-test p-value = 0.250), and none of the individual pre-treatment estimates is statistically significant at conventional levels.

The results in Figure 2 indicate that the policy influences the operational activity of firms in treated provinces. The effect is small in the implementation year (2020), consistent with the policy's effective start date of October 2020, which limited firms' exposure to the subsidy only to the last three months of the year and is therefore unlikely to have altered activity outcomes measured on an annual basis.

From 2021 onward the effect becomes larger and statistically significant, and displays a clear increasing trend. Firms in treated provinces are estimated to be 2.44 percentage points more likely to be active in 2023.

Figure 2: Event study: Activity status



Note: The figure reports the event-study estimates from the Difference-in-Differences specification. The dependent variable is the activity status. The unit of observation is the firm, and firms are not weighted by their 2019 employment level. The specification includes firm and year fixed effects. Standard errors are clustered at the municipality level.

Since activity status is measured as a binary outcome using a linear probability model, the estimated coefficients are directly interpretable as changes in probability, but they do not capture the intensive margin of adjustment, that is whether firms that remain active also change in scale, productivity, or labour input, which is instead addressed in the following sections.

6.2 Input

Figure 3 shows the evolution of firms' labour and capital inputs relative to 2019, the baseline year. The detailed estimates are in the Table A1 in the Appendix.

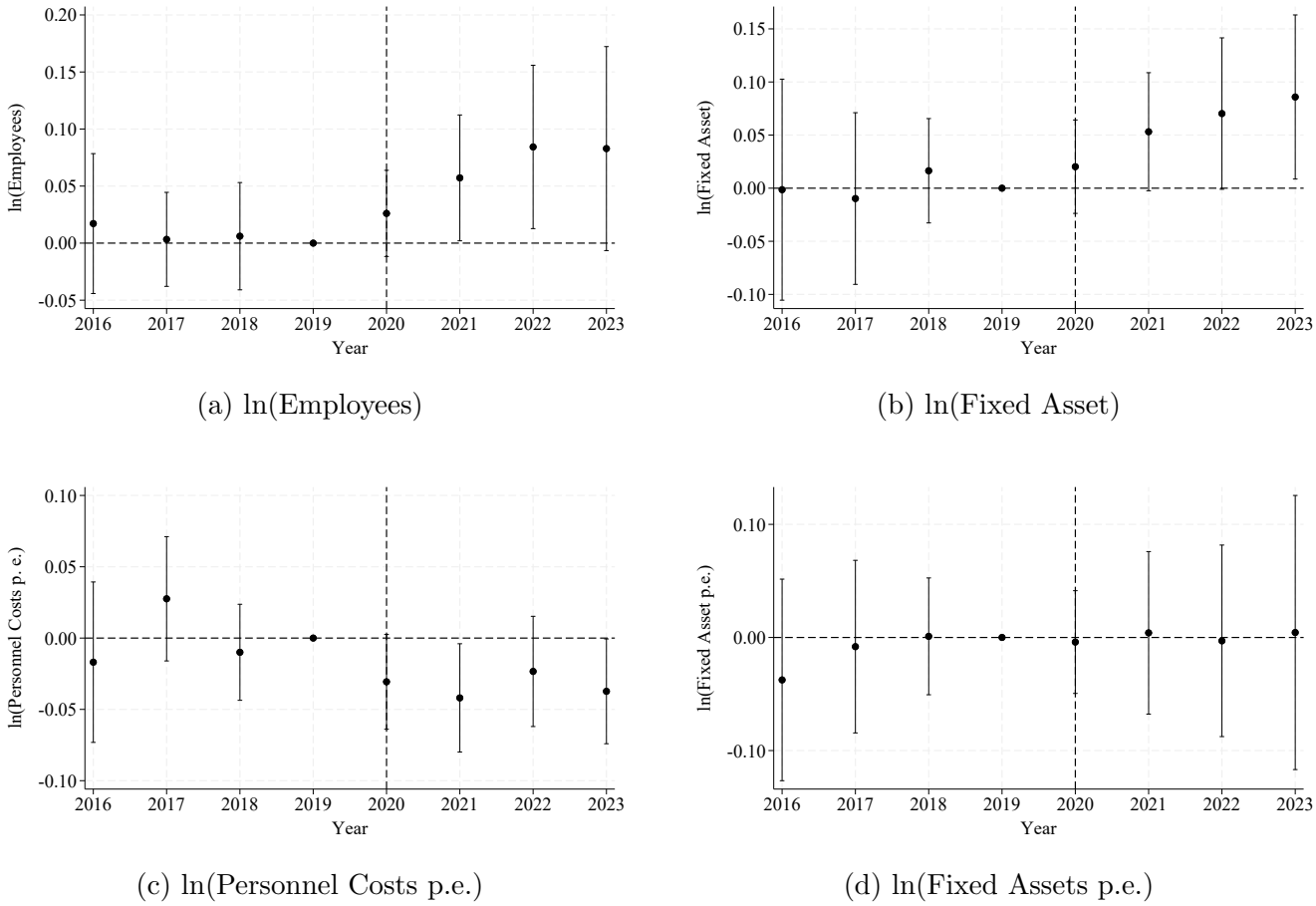
Treated firms display a clear and roughly parallel expansion in both productive factors (Panel a and Panel b). In 2020 the increase is small and not statistically significant, with coefficients of 2.6% for employees and 2.0% for fixed assets. This is consistent with the timing of the policy, which was implemented in October 2020, so that firms were exposed to the treatment for only the last quarter of the year. From 2021 onward the coefficients become statistically significant and grow steadily for both factors, reaching 8.3% for employees and 8.6% for fixed assets by 2023. It is also worth noting that the estimates for the pre-treatment years do not display evidence of diverging trends, supporting the causal interpretation of the estimated effects.

This expansion in employment is not accompanied by a corresponding increase in personnel costs per employee, which instead declines over the same period (Panel c). This pattern – more employees at a lower cost per worker – is consistent with the direct mechanical effect of a payroll tax exemption, and suggests that firms used part of the subsidy to expand their workforce rather than to increase per-worker compensation.

Despite this reduction in the cost of labour, firms show no evidence of substituting capital for labour. The capital-to-labour ratio, measured as fixed assets per employee, remains statistically indistinguishable from zero throughout the post-treatment period (Panel d).

Taken together with the parallel and simultaneous growth in employment and fixed assets described above, this indicates that firms are expanding along the existing production technology. In other words, the input mix remains stable, they are scaling up both inputs proportionally rather than reorganising their productive process.

Figure 3: Event Study – Input



(a) ln(Employees)

(b) ln(Fixed Asset)

(c) ln(Personnel Costs p.e.)

(d) ln(Fixed Assets p.e.)

Note: The figure reports the event-study estimates from the Difference-in-Differences specification. The dependent variables are the logarithm of the number of employees (panel (a)), the logarithm of fixed assets (panel (b)), the logarithm of personnel cost per employee (panel (c)), and the logarithm of fixed assets per employee (panel (d)). The unit of observation is the firm, and firms are weighted by their 2019 employment level. The specification includes firm and year fixed effects. Standard errors are clustered at the municipality level.

6.3 Output

Having established that firms expand their input use, we now turn to output. Figure 4 shows the trends in firms' output, measured as production value (Panel a) and value added (Panel b).

Firms' output variation is close to zero and statistically insignificant in 2020, again consistent with firms' limited exposure to the subsidy in its first, partial year of implementation. From 2021 onward, both measures display a steady upward trend, reaching a statistically significant variation of 9.8% for production value and 11.3% for value added relative to the baseline year by 2023.

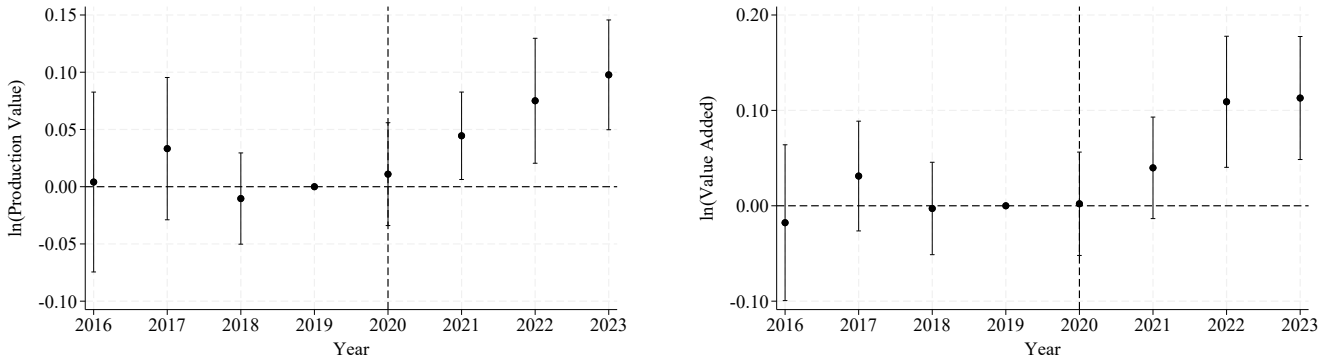
The fact that value added grows at least as fast as gross production value by the end of the sample period suggests that the increase in output is not merely driven by higher intermediate input

purchases, but reflects a genuine increase in the firm's own economic contribution.

However, the pre-treatment estimates provide only marginal support for the parallel trends assumption, with the F-tests only narrowly satisfied. This is visible clearly in Table A2 in the Appendix in which are reported the complete results.

Overall, the expansion in labour and capital documented in the previous section translates coherently into higher output: firms that scale up their productive inputs also produce and generate more value, consistent with a movement along the expansion path rather than a mere reallocation of existing resources.

Figure 4: Event Study – Output



(a) ln(Production Value)

(b) ln(Value Added)

Note: The figure reports the event-study estimates from the Difference-in-Differences specification. The dependent variables are the logarithm of production value (panel (a)) and the logarithm of value added (panel (b)). The unit of observation is the firm, and firms are weighted by their 2019 employment level. The specification includes firm and year fixed effects. Standard errors are clustered at the municipality level.

6.4 Productivity

The only productivity measure available is value added per employee. As shown in Figure 5, the estimated variation is small and never statistically significant across the observed period, ranging between -3.2% and +3.0%. This means that the scale enlargement detected in the two previous sections does not involve any efficiency improvement: firms are producing more with more inputs, but not more efficiently per worker. The absence of a productivity gain is further corroborated by worker-side income data from the MEF archive, discussed in Section 8.1.

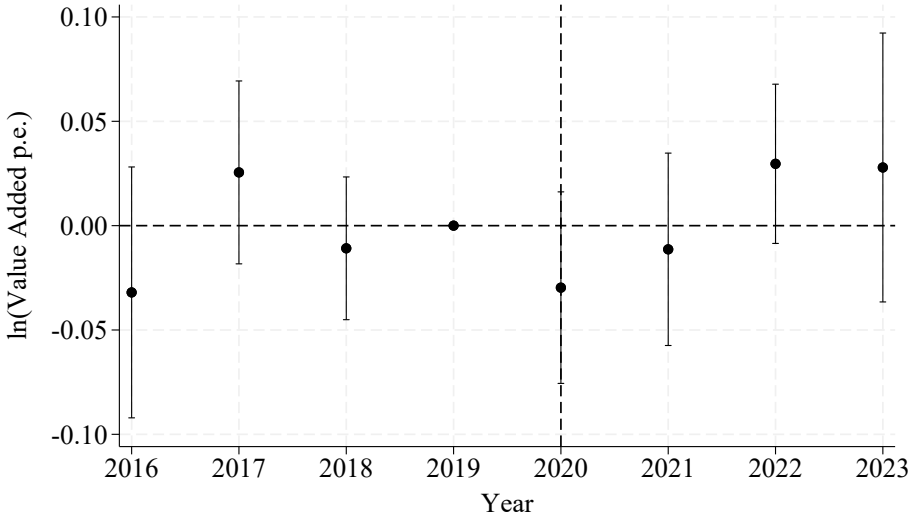
This finding is numerically consistent with the results documented above: by 2023, the estimated growth in value added (11.3%) and in employment (8.3%) differ by roughly the magnitude of the point estimate for value added per employee in the same year (2.8%), even though the latter is not

itself statistically significant. This internal consistency across outcomes reinforces the credibility of the estimated effects.

In Table A2 in the Appendix, it should be noted that the parallel trends assumption is not fully satisfied for this outcome (F-test p = 0.049), which represents a limitation to keep in mind when interpreting these results, similarly to the caveats already raised for other outcomes in this analysis.

Taken together, the results in this and the previous two sections support a coherent interpretation: treated firms expand along their existing production technology, scaling up labour and capital inputs and output in tandem, without any accompanying gain in total factor productivity or labour efficiency.

Figure 5: Event study – Value Added p.e.



Note: The figure reports the event-study estimates from the Difference-in-Differences specification. The dependent variable is the value added per employee. The unit of observation is the firm, and firms are weighted by their 2019 employment level. The specification includes firm and year fixed effects. Standard errors are clustered at the municipality level.

6.5 Profitability

Having shown that firms expand in scale without a corresponding gain in labour productivity, we now examine whether this expansion translates into higher profitability. Figure 6 shows the trends in profit/loss (Panel a) and ROA (Panel b).

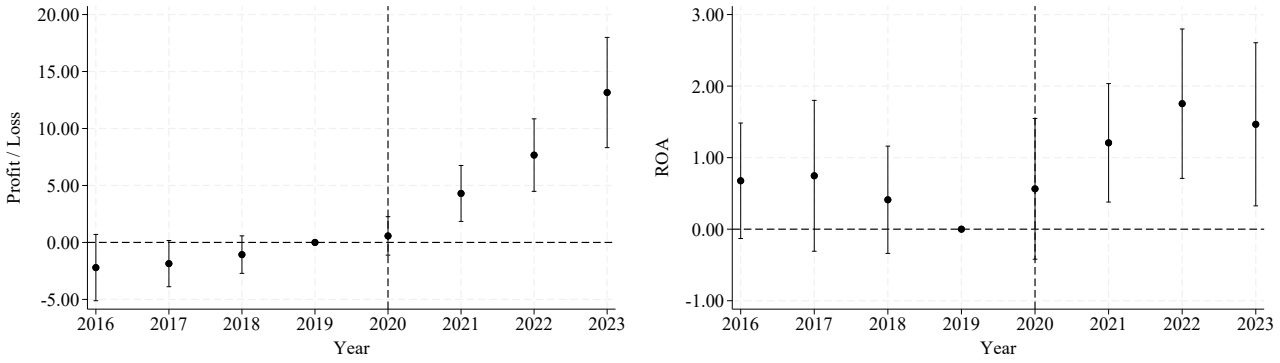
Both profit/loss and ROA display a clear and increasing effect from 2021 onward. Profit/loss rises from a non-significant 0.573 in 2020 to 4.300 in 2021, 7.665 in 2022 and 13.157 by 2023, all statistically significant. ROA follows a similar pattern, growing from an insignificant estimate at t=0 to 1.207, 1.754 and 1.466 percentage points over the same three years.

The pre-treatment estimates for both variables do not point to diverging trends (F-test p-values of 0.307 and 0.409, respectively), supporting a causal interpretation of these effects.

The remaining profitability measures in Table A3 and Figure A2 in the Appendix point in the same direction. ROS increases substantially and significantly throughout the post-treatment period, while ROE rises significantly in 2021 and 2022 but loses statistical significance by 2023, a comparatively weaker pattern than the one observed for profit/loss and ROA. Revenue also grows significantly from 2021 onward, reaching a 9.2% increase by 2023; however, as already noted, the parallel trends assumption for this outcome is not fully satisfied (p=0.023), and this caveat extends, at least in part, to ROS, which is constructed using revenue as its denominator.

Taken together with the results of the previous sections, this pattern is consistent with the interpretation developed throughout this analysis: treated firms expand along their existing production technology, increasing labour, capital and output in tandem without any gain in labour productivity. The additional profitability documented here is therefore unlikely to stem from efficiency gains, and is instead consistent, at least in part, with the mechanical reduction in labour costs induced by the design of the tax cut, as discussed in Section 6.2.

Figure 6: Event Study – Profitability



(a) Profit / Loss (b) ROA

Note: The figure reports the event-study estimates from the Difference-in-Differences specification. The dependent variables are the profit or loss (panel (a)) and the return on assets (panel (b)). The unit of observation is the firm, and firms are weighted by their 2019 employment level for ROA, while they are not for profit or loss. The specification includes firm and year fixed effects. Standard errors are clustered at the municipality level.

7 Robustness Checks

7.1 Spatial Donut Design

As a robustness check on the border-pair identification strategy, the adjacent control provinces is replaced with non-bordering provinces which are: Fermo, Macerata, Perugia, Terni, and Viterbo. This donut design is intended to reduce the risk that treatment effects spill over into bordering control areas, through commuting or the relocation of firms seeking to benefit from the tax cut, thereby threatening SUTVA[1]. Moving the control group away from the border reduces the scope for this contamination, at the cost of weaker local comparability between treated and control units.

Looking at the activity status even though Table A6 (in the Appendix) shows even higher post-treatment estimates with respect to the baseline model, this pattern cannot be attributed to Decontribuzione Sud because treated and the donut control group are significantly different in the pre-treatment period (F-test p = 0.001).

On the input side (Table A6 in the Appendix), employment coefficients remain positive (2.6–4.2%) but are no longer statistically distinguishable from zero. Also the personnel cost per employee show no significant effect in either period. Therefore in the donut model the labour channel loses statistical significance on both variables. The increase in fixed capital, by contrast, is not only confirmed but is even larger, reaching 12.4% by 2023. Firms in the treated group therefore continue to expand input use, but the expansion is now driven by capital rather than both factors, a pattern confirmed by the capital-to-labour ratio (fixed assets per employee), which shows no effect in the baseline model but here it rises significantly in the last two observed years, respectively by 6.4% in 2022 and 6.9% in 2023.

On the output side (Table A7 in the Appendix), production value and value added fail the parallel trends assumption (F-test p = 0.000 in each case), indicating that treated and donut control provinces were not on comparable trajectories even before treatment. The corresponding post-treatment coefficients, though large and highly significant, cannot therefore be interpreted causally and are set aside in the discussion below.

Value added per employee rises in 2020 but shows no significant effect in any subsequent year (Table A7 in the Appendix). Since the policy was only active from the last quarter of 2020 it is plausible that the 2020 coefficient reflects data noise around the implementation date rather than a genuine causal effect.

[1]The Stable Unit Treatment Value Assumption (SUTVA) requires that the potential outcomes of any unit do not vary with the treatment assigned to other units; in a spatial setting, this implies the absence of cross-border spillover effects.

The impact on profitability is short-lived and heterogeneous across measures (Table A8 in the Appendix). Profit/loss and ROE are significant only in 2020, while ROS and ROA remain significant through 2021 before fading in later years. Revenue also fails the parallel trends assumption (F-test $p = 0.000$), so its large and significant post-treatment coefficients cannot be interpreted causally either.

Overall, the donut model confirms the main interpretation of a scale effect without productivity gains. The notable difference from the baseline is that the employment effect documented in Section 6.2 loses statistical significance throughout the post-treatment period. This pattern can indicate that rather than a net job creation the policy induces a reallocation of workers across the border where firms can exploit the fiscal advantage. Moreover, this is consistent also with a simple loss of statistical precision arising from the reduced local comparability of the donut control group. The present design does not allow these two explanations to be distinguished, and this ambiguity is acknowledged as a limitation of the employment result rather than resolved in either direction.

7.2 Exclusion of Construction Sector

In the second robustness check, the baseline model is repeated on a restricted sample that excludes the construction sector (NACE Rev. 2, Section F). This analysis seeks to isolate the causal effect of Decontribuzione Sud by controlling for the confounding impacts of two major concurrent macroeconomic interventions: the Superbonus 110% and the National Recovery and Resilience Plan (PNRR). The former, introduced in 2020, is a 110% (until December 31, 2023) tax credit for the energy efficiency and seismic retrofitting of privately-owned residential buildings. The latter, initiated in 2021, funded large-scale investments across multiple sectors—including green transition, energy efficiency, and transport infrastructure—with a legal requirement that at least 40% of the resources be allocated to Southern regions (which account for roughly 34.4% of the national population, Istat 2026). Because both policies triggered an asymmetric construction boom, excluding this industry mitigates the bulk of these confounding shocks, yielding a cleaner identification of the payroll tax cut's effectiveness across the rest of the economy.

The estimates shown in Table A9 (in the Appendix) confirm the positive effect of the intervention on firms' survival probability, with coefficients slightly lower than those in the baseline method (0.0226 vs 0.0244, in 2023).

The results for the input variables, reported in Table A9 in the Appendix, largely confirm the core findings. On the labour side, the employment and personnel-cost estimates remain similar in magnitude to the baseline and are statistically significant for nearly all post-treatment years. Conversely, while the coefficients for fixed assets maintain a comparable magnitude, they lose

statistical significance except in the final period. Despite this, the capital-to-labour ratio (fixed assets per employee) remains unaffected, with point estimates statistically indistinguishable from zero throughout the observed window.

Output variables—namely, production value and value added (Table A10 in the Appendix)—yield positive coefficients that are slightly smaller than the baseline, reflecting a weaker response in capital accumulation. However, both variables fail the pre-treatment F-test ($p = 0.016$ and $p = 0.048$, respectively). Nonetheless, the overall expansionary trend across both inputs and outputs remains evident in this restricted specification.

Consistent with the main analysis, the policy's effect on firm productivity, measured as value added per employee (Table A10 in the Appendix), remains statistically insignificant.

Finally, profitability outcomes (Table A11 in the Appendix) follow a trajectory highly comparable to the results presented in Section 6.5. The estimates are initially non-significant at the time of treatment but grow to become positive and statistically significant in the subsequent years.

Overall the results demonstrate that Decontribuzione Sud induce a growth in firms' profitability which is determined by the savings on the labour cost, since the growth in inputs and outputs is not accompanied by an efficiency enhancement.

8 Additional Analyses

8.1 Municipality-level Evidence

This section presents the results obtained from the MEF personal income tax data, aggregated at the municipality level.

The analysis considers both the extensive and intensive margins of employment. The extensive margin refers to changes in the number of employed individuals, capturing whether the policy leads to the creation (or formalization) of jobs. The intensive margin refers to changes in the aggregate income received by workers, which may reflect higher wages, longer hours, or a shift towards better-paid positions.

The empirical model is substantially similar to the one described in Section 5, with a few adjustments. The unit of observation is now the municipality, and observations are weighted by their 2019 baseline level for the respective outcome (e.g., when estimating the number of employees or employment income, municipalities are weighted by their 2019 employment level). Moreover, the specification includes municipality fixed effects rather than firm fixed effects.

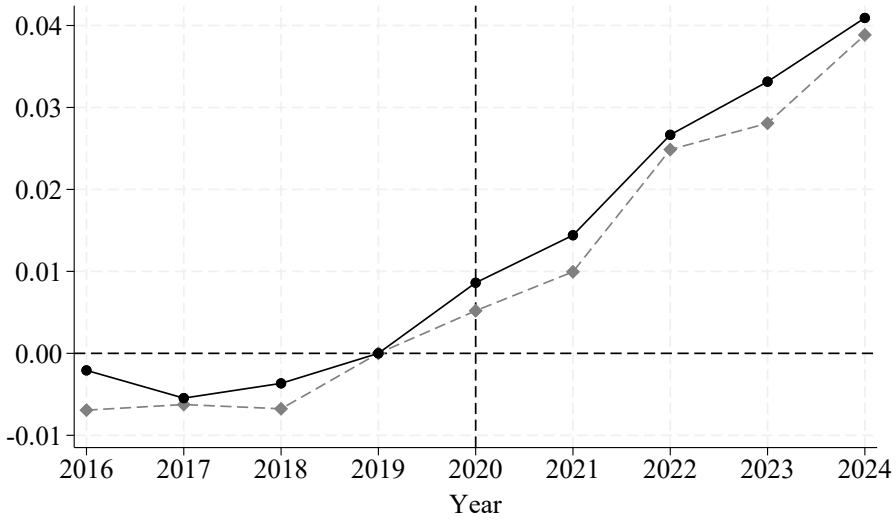
Aggregating at the municipality level has an important advantage: it captures the total effect of the policy on the local labour market, rather than only its effect on the firms observed in the balance-sheet data. First, the MEF data cover all sources of labour income, including self-employment, which is not observed in firm-level data. Second, aggregation internalises spillovers between firms, such as workers reallocating from non-eligible to eligible firms. Third, aggregate outcomes reflect firm entry and exit, since jobs created by new firms and jobs lost through closures are both counted, whereas firm-level analyses condition on surviving firms.

Table A4 in the Appendix reports the results for the extensive margin. All dimensions, except for the number of self-employed individuals, show positive and significant post-treatment coefficients, confirming the increase in employment already documented using the AIDA data. However, for the number of taxpayers and employees, the pre-treatment F-test rejects the null hypothesis of parallel trends (respectively $p = 0.018$ and $p = 0.039$). Therefore, it is not possible to rule out that part of the estimated post-treatment effect reflects a pre-existing divergence between treated and control municipalities.

A similar pattern emerges for the intensive margin, reported in Table A5 in the Appendix. All outcomes—again, with the exception of self-employment—display a significant increase throughout the post-treatment period. However, in this case, the pre-treatment F-test for taxable income provides evidence against parallel trends (p-value = 0.017).

As can be seen in Figure 7, the post-treatment coefficients for the number of employees (0.005–0.039) and for aggregate employment income (0.009–0.041) follow a comparable trajectory and are very close in magnitude. This indicates that the increase in aggregate employment income is almost entirely explained by the rise in the number of employees, rather than by a growth in average income per worker. This finding reinforces the results obtained with the AIDA data, where we document a scale effect: firms expand both labour and capital inputs but experience no significant improvement in productivity, as measured by value added per employee. The absence of a productivity gain is consistent with, and helps explain, the observed stability of average employee income.

Figure 7: Event study: Number of employees and employment income



Note: The figure reports the event-study estimates from the Difference-in-Differences specification. The continuous black line is the employment income, while the dashed grey line is the number of employees. The unit of observation is the municipality, and municipalities are weighted by their number of employees in 2019. The specification includes municipality and year fixed effects. Standard errors are clustered at the municipality level.

The fact that the number of individuals paying a net tax grows faster, in relative terms, than the number of taxpayers with positive taxable income is unlikely to reflect a genuine increase in average income; rather, it points to a purely compositional effect. New workers entering formal employment—driven by the extensive-margin effect documented above—typically earn wages above the relatively low no-tax area threshold. Consequently, a similar absolute increase in individuals translates into a larger percentage increase for the smaller baseline of net taxpayers compared to the larger baseline of individuals with positive taxable income. While this interpretation cannot be directly verified with municipality-level aggregate data, it does not require assuming any increase in average wages, perfectly aligning with the extensive-margin dynamics highlighted throughout this section.

8.2 Cost-Benefit Analysis

Beyond estimating the causal effect of the policy, it can be useful to assess its costs relative to the benefits generated. In this section, the focus is still on those municipalities located in provinces in Southern Italy bordering Central Italy, consistent with the estimates presented above.

This cost-benefit analysis relies on the simplifying assumption that the sole objective of the intervention was to increase employment. To this end, the fiscal cost of Decontribuzione Sud is set against the estimated employment gains. In particular, the analysis presents three different

indices, each capturing a slightly different notion of "benefit". The indices share the same numerator calculated by Scrutinio (2026), who reconstructs the fiscal cost of the exemption directly from UNIEMENS[2] payroll records, obtaining an expenditure of €1.44 billion for provinces close to the policy border.

The first measure relates the fiscal outlay to the estimated number of additional employees generated by the policy. Multiplying the cumulative percentage effect on employment reported in Table A4 by the 2019 employment level in the treated provinces (534,502) yields 57,157 additional worker-years over the sample period. Dividing the €1.44 billion cost by this number gives an implied cost of €25,194 per worker-year generated.

The second index substitutes the number of employees with the number of taxpayers in the denominator. Applying the same procedure—cumulative estimated effect times the 2019 baseline, here 993,307 taxpayers—produces 51,337 additional taxpayer-years, and a cost of €28,050 per taxpayer-year.

The third measure values the additional labour income generated by the policy. Applying the coefficients estimated for employment income (Table A5) to the 2019 baseline (approximately €928.2 million) implies a cumulative gain of €1.15 billion in aggregate employment income, so that each additional euro of labour income cost approximately €1.25 in public expenditure.

The cost per additional taxpayer-year is higher than the cost per additional employee-year, despite the taxpayer baseline being almost twice as large. This is consistent with the number of taxpayers being a considerably broader concept than the number of employees: it includes individuals earning income from other sources such as self-employment and pensions. Part of the estimated increase in dependent employment plausibly reflects workers reclassified from other income categories (e.g., self-employment) into dependent employment, rather than individuals newly entering the tax base altogether; such a shift raises the employee count without a corresponding increase in the taxpayer count. Taken together, these compositional factors make the taxpayer-based index a less precise, and likely upwardly biased, measure of the fiscal cost of job creation relative to the employee-based index.

The magnitude of these indices differs from those reported in Scrutinio (2026), who estimates a cost of approximately €19,700 per additional worker-year. Since the numerator is identical across the two exercises (the same €1.44 billion estimate is used), the divergence is entirely attributable to differences in the denominator. UNIEMENS records dependent employment relationships in the private sector directly at the place of work, whereas the MEF data used here are assigned by place

[2]UNIEMENS is the INPS administrative database that collects all monthly information submitted by employers regarding employee wages, social security contributions, and employment statuses.

of residence and are based on annual tax returns rather than administrative payroll records. The 2019 baseline employment level used here (534,502) is larger than that used in Scrutinio (2026) (311,000), consistent with a broader and less policy-specific population being captured by the MEF definition of "dependent employment and assimilated income". A larger, less targeted baseline population tends to be associated with a smaller estimated percentage effect, and therefore, for a fixed fiscal cost, with a higher cost per unit of benefit.

9 Concluding Remarks

Introduced in 2020 to mitigate the economic crisis of the COVID-19 pandemic, Decontribuzione Sud reduced employers' social security contributions in Southern Italy to support employment and firm activity in the country's least developed regions. This thesis evaluated the policy's effects using a Difference-in-Differences event study design, comparing firms in treated provinces with untreated counterparts just across the policy border. Drawing on firm-level balance-sheet data (AIDA) and municipal income tax data (MEF) over the 2016–2024 period, the analysis covered a broad set of outcomes, spanning firm survival, employment, labour costs, capital accumulation, productivity, and profitability.

The results demonstrate that Decontribuzione Sud is effective in increasing the survival probability of firms and boosting employment but the aggregated data highlight a weak impact on wages. Rather than inducing a substitution between labour and capital, the intervention generates a proportional scale effect, allowing firms to expand overall output and profitability. However, this rise in profitability comes primarily from the mechanical labour cost savings generated by the tax cut, rather than from underlying improvements in production efficiency.

This mechanism contrasts with the recent evaluation by Albanese et al. (2026), who find no significant effect on employment, wages, or fixed assets, but detect an increase in value added per worker and other profitability metrics. A plausible explanation for this divergence is their restricted focus on small and medium-sized enterprises (SMEs), suggesting that the employment and scale effects detected in this thesis may be primarily driven by larger firms. Future research could empirically test this hypothesis by examining outcomes from 2025 onward, when the subsidy became exclusively restricted to SMEs.

Despite these robust findings, this analysis faces some limitations. First, one of the policy's objectives, clearly stated in the 2025 legislative update, is to promote stable employment through open-ended contracts. However, the data used in this analysis cannot distinguish the specific types of contracts signed as a result of the intervention. Second, there is a limitation concerning

data completeness: the variable *Employees*, which is among the most informative and is used to construct all per-worker variables, suffers from a high share of missing values in the last observed year (approximately 25.7% of the records). Finally, over the period of analysis, the effects of Decontribuzione Sud may overlap with those of other concurrent policies, most notably the National Recovery and Resilience Plan (PNRR) and the Superbonus. While this concern is partly addressed by a robustness check excluding the construction sector, it cannot be ruled out that the residual influence of these policies partially affects the estimates.

Ultimately, these findings offer valuable insights for the design of future cohesion policies. The cost-benefit analysis highlights that while broad payroll tax cuts can be effective instruments for safeguarding businesses and stimulating job creation, they involve a substantial fiscal burden. The estimates indicate a cost of approximately €25,200 per additional worker-year, meaning that each extra euro of labour income required roughly €1.25 in public expenditure. Furthermore, as the firm-level data show, these measures generally fall short of enhancing underlying productivity. Therefore, to achieve long-term regional convergence and maximize the efficiency of public spending, such expensive cost-saving measures likely need to be coupled with targeted investments in innovation, infrastructure, and human capital.

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10 Appendix

Figure A1: Provinces Included in the Analysis



Note: The map displays the provinces used in the empirical analysis. Provinces in orange belong to the treated group, provinces in yellow belong to the control group, and the area in sky blue represent the control provinces in the donut design.

Table A1: Event study – Activity and Input

	Active Status	ln(Employees)	ln(Personnel Costs p. e.)	ln(Fixed Asset)	ln(Fixed Asset p.e.)
Pre-treatment period					
t = -4 (2016)	0.0034	0.017	−0.017	−0.001	−0.038
	(0.0052)	(0.031)	(0.029)	(0.053)	(0.045)
t = -3 (2017)	0.0037	0.003	0.028	−0.010	−0.008
	(0.0039)	(0.021)	(0.022)	(0.041)	(0.039)
t = -2 (2018)	−0.0022	0.006	−0.010	0.016	0.001
	(0.0034)	(0.024)	(0.017)	(0.025)	(0.026)
Post-treatment period					
t = 0 (2020)	0.0034	0.026	−0.031*	0.020	−0.004
	(0.0032)	(0.019)	(0.017)	(0.022)	(0.023)
t = +1 (2021)	0.0154***	0.057**	−0.042**	0.053*	0.004
	(0.0039)	(0.028)	(0.019)	(0.028)	(0.037)
t = +2 (2022)	0.0206***	0.084**	−0.023	0.070*	−0.003
	(0.0041)	(0.037)	(0.020)	(0.036)	(0.043)
t = +3 (2023)	0.0244***	0.083*	−0.037**	0.086**	0.004
	(0.0043)	(0.046)	(0.019)	(0.039)	(0.062)
Observations	293885	166003	163708	175103	158975
R-squared	0.762	0.963	0.779	0.959	0.896
Parallel trends F-test (p-value)	0.250	0.906	0.084	0.663	0.623

Note: The table reports the event-study estimates from the baseline Difference-in-Differences model. The unit of observation is the firm. Active status is a linear probability model; all other outcomes are in logarithms. All specifications include firm and year fixed effects and are weighted by 2019 employment, except for active status which is unweighted. Standard errors clustered at municipality levels in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A2: Event study – Output and Productivity

	ln(Production Value)	ln(Value Added)	ln(Value Added p.e.)
Pre-treatment period			
t = -4 (2016)	0.004	−0.018	−0.032
	(0.040)	(0.042)	(0.031)
t = -3 (2017)	0.033	0.031	0.026
	(0.032)	(0.029)	(0.022)
t = -2 (2018)	−0.010	−0.003	−0.011
	(0.020)	(0.025)	(0.017)
Post-treatment period			
t = 0 (2020)	0.011	0.002	−0.030
	(0.023)	(0.028)	(0.023)
t = +1 (2021)	0.044**	0.040	−0.011
	(0.019)	(0.027)	(0.024)
t = +2 (2022)	0.075***	0.109***	0.030
	(0.028)	(0.035)	(0.019)
t = +3 (2023)	0.098***	0.113***	0.028
	(0.024)	(0.033)	(0.033)
Observations	181714	173943	160106
R-squared	0.956	0.955	0.784
Parallel trends F-test (p-value)	0.052	0.092	0.049

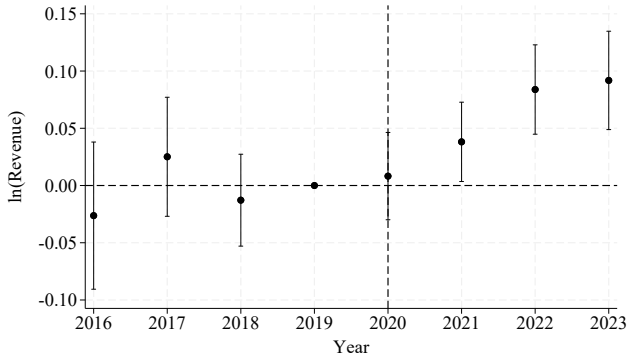
Note: The table reports the event-study estimates from the baseline Difference-in-Differences model. The unit of observation is the firm. All outcomes are in logarithms. All specifications include firm and year fixed effects and are weighted by 2019 employment. Standard errors clustered at municipality levels in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Event study – Profitability

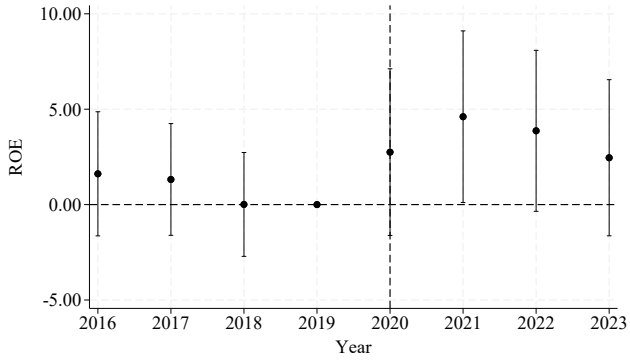
	ln(Revenue)	Profit / Loss	ROS	ROE	ROA
Pre-treatment period					
t = -4 (2016)	−0.026	−2.206	0.001	1.613	0.677
	(0.033)	(1.484)	(0.294)	(1.660)	(0.412)
t = -3 (2017)	0.025	−1.859*	0.022	1.317	0.746
	(0.027)	(1.035)	(0.295)	(1.494)	(0.538)
t = -2 (2018)	−0.013	−1.068	0.090	0.009	0.411
	(0.020)	(0.840)	(0.272)	(1.389)	(0.383)
Post-treatment period					
t = 0 (2020)	0.008	0.573	0.012	2.750	0.564
	(0.019)	(0.860)	(0.346)	(2.229)	(0.502)
t = +1 (2021)	0.038**	4.300***	1.173***	4.606**	1.207***
	(0.018)	(1.252)	(0.405)	(2.295)	(0.422)
t = +2 (2022)	0.084***	7.665***	1.453***	3.868*	1.754***
	(0.020)	(1.624)	(0.480)	(2.153)	(0.533)
t = +3 (2023)	0.092***	13.157***	0.733**	2.455	1.466**
	(0.022)	(2.468)	(0.309)	(2.088)	(0.582)
Observations	179395	267506	167826	167931	184601
R-squared	0.962	0.640	0.492	0.386	0.440
Parallel trends F-test (p-value)	0.023	0.307	0.987	0.441	0.409

Note: The table reports the event-study estimates from the baseline Difference-in-Differences model. The unit of observation is the firm. Revenue is in logarithms, Profit/Loss is in levels, while ROS, ROE, and ROA are in percentage. Profit/Loss, ROS, ROE and ROA are winsorised at 1%. All specifications include firm and year fixed effects and are weighted by 2019 employment, except for revenue which is unweighted. Standard errors clustered at municipality levels in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

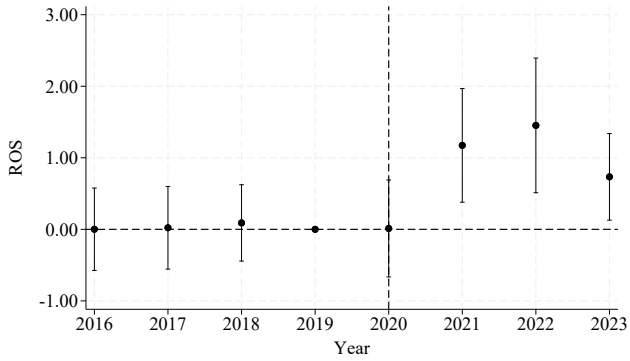
Figure A2: Event Study – Profitability



(a) ln(Revenue)



(b) ROE



(c) ROS

Note: The figure reports the event-study estimates from the Difference-in-Differences specification. The dependent variables are logarithm of revenues (panel (a)), the return on equity (panel (b)), and the return on sales (panel (c)). The unit of observation is the firm, and firms are weighted by their 2019 employment level. The specification includes firm and year fixed effects. Standard errors are clustered at the municipality level.

Table A4: Municipality-level Estimates – Extensive margin

	Num. Taxpayers	Num. Taxable Income	Num. Net Tax	Num. Employment Income	Num. Self-employment Income
Pre-treatment period					
t = -4 (2016)	−0.002	−0.000	0.005	−0.007	0.009
	(0.003)	(0.003)	(0.004)	(0.005)	(0.020)
t = -3 (2017)	−0.003	−0.002	0.000	−0.006*	−0.011
	(0.003)	(0.003)	(0.004)	(0.004)	(0.019)
t = -2 (2018)	−0.005***	−0.003	−0.002	−0.007***	−0.011
	(0.002)	(0.002)	(0.003)	(0.002)	(0.016)
Post-treatment period					
t = 0 (2020)	0.002	0.002	0.002	0.005**	−0.025
	(0.002)	(0.002)	(0.002)	(0.002)	(0.019)
t = +1 (2021)	0.003	0.004*	0.009***	0.010***	−0.034
	(0.002)	(0.002)	(0.003)	(0.003)	(0.021)
t = +2 (2022)	0.011***	0.013***	0.024***	0.025***	−0.022
	(0.003)	(0.003)	(0.005)	(0.005)	(0.021)
t = +3 (2023)	0.014***	0.016***	0.034***	0.028***	−0.028
	(0.004)	(0.004)	(0.007)	(0.006)	(0.023)
t = +4 (2024)	0.020***	0.023***	0.041***	0.039***	−0.014
	(0.005)	(0.005)	(0.008)	(0.007)	(0.025)
Observations	4869	4869	4869	4869	3435
R-squared	1.000	1.000	1.000	1.000	0.997
Parallel trends F-test (p-value)	0.018	0.261	0.006	0.039	0.045
Mean (Th)	3.622	3.444	2.582	1.955	0.063

Note: The table reports the event-study estimates from the Difference-in-Differences model, estimated on MEF personal income tax data. The unit of observation is the municipality. The outcomes are the logarithms of the number of individuals in each category. All specifications include municipality and year fixed effects, and municipalities are weighted by their 2019 value of the respective outcome. Standard errors clustered at municipality level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Municipality-level Estimates – Intensive margin

	Taxable Income	Net Tax	Employment Income	Self-employment Income
Pre-treatment period				
t = -4 (2016)	−0.000	−0.009	−0.002	0.031
	(0.005)	(0.008)	(0.006)	(0.026)
t = -3 (2017)	−0.004	−0.010	−0.005	0.023
	(0.004)	(0.006)	(0.004)	(0.020)
t = -2 (2018)	−0.004	−0.006	−0.004	0.019
	(0.003)	(0.004)	(0.003)	(0.015)
Post-treatment period				
t = 0 (2020)	0.004**	0.004	0.009***	−0.010
	(0.002)	(0.003)	(0.003)	(0.017)
t = +1 (2021)	0.008***	0.011***	0.014***	−0.048**
	(0.003)	(0.004)	(0.004)	(0.020)
t = +2 (2022)	0.019***	0.029***	0.027***	0.015
	(0.004)	(0.005)	(0.006)	(0.023)
t = +3 (2023)	0.025***	0.038***	0.033***	0.032
	(0.005)	(0.007)	(0.007)	(0.032)
t = +4 (2024)	0.031***	0.046***	0.041***	0.080***
	(0.007)	(0.008)	(0.009)	(0.027)
Observations	4869	4869	4869	3435
R-squared	1.000	0.999	0.999	0.993
Parallel trends F-test (p-value)	0.017	0.192	0.125	0.634
Mean (Mln)	63.817	11.154	36.316	27.460

Note: The table reports the event-study estimates from the Difference-in-Differences model, estimated on MEF personal income tax data. The unit of observation is the municipality. The outcomes are the logarithms of the aggregate amounts declared in each category. All specifications include municipality and year fixed effects, and municipalities are weighted by their 2019 value of the respective outcome. Standard errors clustered at municipality level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Donut Event study – Activity and Input

	Active Status	ln(Employees)	ln(Personnel Costs p. e.)	ln(Fixed Asset)	ln(Fixed Asset p.e.)
Pre-treatment period					
t = -4 (2016)	−0.0103**	−0.044*	−0.053*	−0.055	0.001
	(0.0051)	(0.026)	(0.029)	(0.035)	(0.041)
t = -3 (2017)	−0.0077**	−0.006	−0.026	0.017	0.042
	(0.0037)	(0.030)	(0.028)	(0.048)	(0.046)
t = -2 (2018)	−0.0123***	−0.021	−0.023	0.000	0.015
	(0.0030)	(0.024)	(0.024)	(0.017)	(0.031)
Post-treatment period					
t = 0 (2020)	0.0074**	0.026	−0.010	0.014	−0.010
	(0.0031)	(0.025)	(0.025)	(0.018)	(0.028)
t = +1 (2021)	0.0249***	0.042	−0.027	0.067**	0.032
	(0.0038)	(0.028)	(0.024)	(0.026)	(0.030)
t = +2 (2022)	0.0292***	0.041	0.027	0.104***	0.064*
	(0.0042)	(0.027)	(0.026)	(0.032)	(0.037)
t = +3 (2023)	0.0293***	0.030	−0.002	0.124***	0.069*
	(0.0050)	(0.027)	(0.022)	(0.030)	(0.041)
Observations	315881	175862	173152	185443	169969
R-squared	0.789	0.969	0.775	0.961	0.898
Parallel trends F-test (p-value)	0.001	0.161	0.231	0.201	0.564

Note: The table reports the event-study estimates from the donut Difference-in-Differences model. The unit of observation is the firm. Active status is a linear probability model; all other outcomes are in logarithms. All specifications include firm and year fixed effects and are weighted by 2019 employment, except for active status which is unweighted. Standard errors clustered at municipality levels in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Donut Event study – Output and Productivity

	ln(Production Value)	ln(Value Added)	ln(Value Added p.e.)
Pre-treatment period			
t = -4 (2016)	−0.097***	−0.113***	−0.056*
	(0.027)	(0.026)	(0.031)
t = -3 (2017)	−0.041*	−0.041**	−0.032
	(0.021)	(0.020)	(0.031)
t = -2 (2018)	−0.039***	−0.046***	−0.023
	(0.011)	(0.012)	(0.026)
Post-treatment period			
t = 0 (2020)	0.059***	0.085***	0.063**
	(0.017)	(0.025)	(0.028)
t = +1 (2021)	0.061***	0.052**	0.017
	(0.017)	(0.023)	(0.029)
t = +2 (2022)	0.076**	0.088**	0.044
	(0.033)	(0.037)	(0.037)
t = +3 (2023)	0.101***	0.079***	0.032
	(0.023)	(0.021)	(0.027)
Observations	191002	182870	169653
R-squared	0.961	0.957	0.765
Parallel trends F-test (p-value)	0.000	0.000	0.232

Note: The table reports the event-study estimates from the donut Difference-in-Differences model. The unit of observation is the firm. All outcomes are in logarithms. All specifications include firm and year fixed effects and are weighted by 2019 employment. Standard errors clustered at municipality levels in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Donut Event study – Profitability

	ln(Revenue)	Profit / Loss	ROS	ROE	ROA
Pre-treatment period					
t = -4 (2016)	−0.096***	2.520	−0.248	0.668	−0.153
	(0.027)	(1.868)	(0.312)	(1.355)	(0.330)
t = -3 (2017)	−0.034*	−1.225	−0.371	−1.070	0.043
	(0.020)	(1.340)	(0.321)	(1.587)	(0.357)
t = -2 (2018)	−0.041***	0.298	−0.394	−0.385	−0.035
	(0.010)	(0.990)	(0.280)	(1.455)	(0.285)
Post-treatment period					
t = 0 (2020)	0.057***	5.521***	1.352***	3.767*	1.364**
	(0.016)	(1.236)	(0.481)	(2.085)	(0.656)
t = +1 (2021)	0.073***	0.844	0.872**	2.413	0.819**
	(0.017)	(1.517)	(0.343)	(2.105)	(0.402)
t = +2 (2022)	0.111***	1.527	0.213	0.723	0.551
	(0.034)	(2.459)	(0.364)	(1.868)	(0.501)
t = +3 (2023)	0.109***	4.918	−0.137	−0.088	0.390
	(0.022)	(3.269)	(0.353)	(1.828)	(0.522)
Observations	188873	287067	176792	176419	193478
R-squared	0.961	0.671	0.504	0.392	0.452
Parallel trends F-test (p-value)	0.000	0.043	0.557	0.488	0.876

Note: The table reports the event-study estimates from the donut Difference-in-Differences model. The unit of observation is the firm. Revenue is in logarithms, Profit/Loss is in levels, while ROS, ROE, and ROA are in percentage. Profit/Loss, ROS, ROE and ROA are winsorised at 1%. All specifications include firm and year fixed effects and are weighted by 2019 employment, except for revenue which is unweighted. Standard errors clustered at municipality levels in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A9: Event Study Excluding the Construction Sector – Activity and Input

	Active Status	ln(Employees)	ln(Personnel Costs p. e.)	ln(Fixed Asset)	ln(Fixed Asset p.e.)
Pre-treatment period					
t = -4 (2016)	0.0032	0.024	−0.024	−0.012	−0.058
	(0.0058)	(0.033)	(0.030)	(0.059)	(0.050)
t = -3 (2017)	0.0021	0.014	0.031	−0.010	−0.017
	(0.0040)	(0.024)	(0.024)	(0.045)	(0.043)
t = -2 (2018)	−0.0033	0.010	−0.009	0.015	−0.005
	(0.0040)	(0.027)	(0.019)	(0.028)	(0.030)
Post-treatment period					
t = 0 (2020)	0.0025	0.033	−0.036*	0.018	−0.013
	(0.0036)	(0.021)	(0.019)	(0.025)	(0.025)
t = +1 (2021)	0.0138***	0.066**	−0.050**	0.048	−0.006
	(0.0043)	(0.031)	(0.023)	(0.031)	(0.041)
t = +2 (2022)	0.0180***	0.084**	−0.023	0.062	−0.005
	(0.0042)	(0.041)	(0.022)	(0.039)	(0.048)
t = +3 (2023)	0.0226***	0.090*	−0.043**	0.073*	−0.011
	(0.0048)	(0.051)	(0.021)	(0.043)	(0.068)
Observations	235653	135453	133369	143370	130389
R-squared	0.759	0.966	0.795	0.961	0.900
Parallel trends F-test (p-value)	0.256	0.901	0.072	0.796	0.449

Note: The table reports the event-study estimates from the Difference-in-Differences excluding the construction sector. The unit of observation is the firm. Active status is a linear probability model; all other outcomes are in logarithms. All specifications include firm and year fixed effects and are weighted by 2019 employment, except for active status which is unweighted. Standard errors clustered at municipality levels in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A10: Event Study Excluding the Construction Sector – Output and Productivity

	ln(Production Value)	ln(Value Added)	ln(Value Added p.e.)
Pre-treatment period			
t = -4 (2016)	0.007	−0.014	−0.036
	(0.042)	(0.045)	(0.032)
t = -3 (2017)	0.045	0.043	0.026
	(0.033)	(0.031)	(0.024)
t = -2 (2018)	−0.010	0.000	−0.013
	(0.022)	(0.027)	(0.019)
Post-treatment period			
t = 0 (2020)	0.008	0.002	−0.039
	(0.026)	(0.031)	(0.026)
t = +1 (2021)	0.036	0.037	−0.020
	(0.023)	(0.031)	(0.027)
t = +2 (2022)	0.065**	0.099**	0.021
	(0.031)	(0.040)	(0.022)
t = +3 (2023)	0.068**	0.094**	0.011
	(0.028)	(0.037)	(0.035)
Observations	148029	141212	130246
R-squared	0.961	0.959	0.799
Parallel trends F-test (p-value)	0.016	0.048	0.073

Note: The table reports the event-study estimates from the Difference-in-Differences model excluding the construction sector. The unit of observation is the firm. All outcomes are in logarithms. All specifications include firm and year fixed effects and are weighted by 2019 employment. Standard errors clustered at municipality levels in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A11: Event Study Excluding the Construction Sector – Profitability

	ln(Revenue)	Profit / Loss	ROS	ROE	ROA
Pre-treatment period					
t = -4 (2016)	−0.034	−2.422	0.078	2.095	0.813*
	(0.035)	(1.569)	(0.316)	(1.863)	(0.459)
t = -3 (2017)	0.028	−2.088*	0.051	1.394	0.864
	(0.028)	(1.208)	(0.328)	(1.692)	(0.606)
t = -2 (2018)	−0.016	−0.989	0.091	−0.086	0.507
	(0.022)	(1.029)	(0.301)	(1.611)	(0.434)
Post-treatment period					
t = 0 (2020)	−0.002	0.603	−0.040	2.988	0.631
	(0.023)	(1.060)	(0.407)	(2.620)	(0.591)
t = +1 (2021)	0.027	3.908**	1.284***	5.178*	1.319***
	(0.020)	(1.564)	(0.464)	(2.671)	(0.486)
t = +2 (2022)	0.072***	6.315***	1.406**	4.058	1.787***
	(0.022)	(1.756)	(0.552)	(2.510)	(0.603)
t = +3 (2023)	0.059***	9.178***	0.676**	2.274	1.470**
	(0.022)	(2.303)	(0.337)	(2.418)	(0.636)
Observations	146223	214684	137461	135290	150341
R-squared	0.967	0.665	0.499	0.384	0.441
Parallel trends F-test (p-value)	0.011	0.309	0.982	0.306	0.353

Note: The table reports the event-study estimates from the Difference-in-Differences model excluding the construction sector. The unit of observation is the firm. Revenue is in logarithms, Profit/Loss is in levels, while ROS, ROE, and ROA are in percentage. Profit/Loss, ROS, ROE and ROA are winsorised at 1%. All specifications include firm and year fixed effects and are weighted by 2019 employment, except for revenue which is unweighted. Standard errors clustered at municipality levels in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.